

Pravar Mishra

Software Engineer · Backend, Data Pipelines & Geospatial Systems

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PROFILE

Software engineer with a Computer Science background (B.Tech CSE, GATE qualified) who builds **backend services, data pipelines and ML systems**, with a focus on satellite and geospatial data. Currently a SAR Image Consultant at **ICEYE**; previously shipped imagery services at **Dhruva Space** and backend authentication and messaging at **NRSC (ISRO)**. Works across Python and Java, from REST APIs and service-to-service messaging to PyTorch models.

TECHNICAL SKILLS

Languages	Python, C++, Java, SQL
Backend & Infra	FastAPI, Django REST, Spring Boot, REST API design, PostgreSQL, Git
ML & Data	PyTorch, CNN training & evaluation, object detection, experiment design & error analysis
Geospatial	Cloud-Optimized GeoTIFF, Titiler, STAC / PGSTAC / stac-fastapi, GeoJSON, shapefiles, QGIS, TLE handling
SAR Domain	Imaging geometry, incidence angle effects, speckle, radar shadow, image-quality assessment, target interpretation

EXPERIENCE

SAR Image Consultant · ICEYE, Espoo, Finland (Remote) Jul 2026 - Present

- Work with operational products from ICEYE's SAR constellation; trained on-site at ICEYE HQ in Espoo in **SAR imaging geometry, product formats and image-quality assessment**.

Software Engineer Intern · Dhruva Space, Hyderabad Jan 2025 - Mar 2025

- Built an **ML object-detection service (CountGD)** that detects and counts objects in satellite imagery, the automated feature-extraction pattern behind vessel and vehicle detection.
- Designed and delivered a **map tile service (Titiler over COGs)** and a **searchable STAC catalogue** backed by PostgreSQL (PGSTAC, stac-fastapi, STAC Browser), serving large imagery archives to end users.
- Implemented satellite **TLE data handling** for orbital and collection planning.

Software Intern · NRSC, ISRO, Hyderabad Sep 2023 - Nov 2023

- Designed a prototype **authentication architecture (Spring Security 6+)** that was adopted for NRSC applications.
- Built **inter-service communication** over HTTP and **RabbitMQ** between backend services at India's national remote-sensing centre.

PROJECTS & RESEARCH

SAR Target Classification on MSTAR · Python, PyTorch · Independent research 2025 - 2026

- Trained a CNN for **10-class military vehicle recognition from SAR imagery**, reaching **96.5% accuracy** on the standard MSTAR split, where test data is collected at a different depression angle than training, a real operational distribution shift.
- **Operationally relevant error analysis**: diagnosed a dominant false-positive class from the confusion matrix and fixed it by capacity scaling (errors **~237** → **~84**); characterised confident-but-wrong failures against physically similar vehicles, the failure mode that matters most in deployment.

EDUCATION

B.Tech, Computer Science & Engineering · Vellore Institute of Technology, Andhra Pradesh 2021 - 2025

- CGPA: 8.75 / 10

GATE (Computer Science), Qualified · All India Rank 6366 · Score 520 2026

ADDITIONAL

NCC 'A' Certificate · English (fluent), Hindi (native)